

WORLD DEVELOPMENT INDICATORS (WDI) PROJECT

GDP AND ECONOMIC GROWTH

TECHNOLOGY AND INTERNET ADOPTION

Problem Statement

The project explores two distinct themes using WDI data:

1. Economic Performance (GDP Growth), measuring the size, growth, and distribution of economies.
2. Technology Adoption (Internet, Mobile, Broadband), assessing digital connectivity across income groups and regions.

The objective is to provide insights into global development patterns and highlight disparities between countries.

Dataset Description

- **Source:** World Bank World Development Indicators (WDI)
- **Indicators:** Seven Indicators were selected. They were;

Indicator	Code
GDP (current US\$)	NY.GDP.MKTP.CD
GDP growth (annual %)	NY.GDP.MKTP.KD.ZG
GDP per capita, PPP	NY.GDP.PCAP.PP.KD
GDP per capita growth (annual %)	NY.GDP.PCAP.KD.ZG
Internet users (% of population)	IT.NET.USER.ZS
Mobile subscriptions (per 100 people)	IT.CEL.SETS.P2
Fixed broadband subscriptions (per 100 people)	IT.NET.BBND.P2

- **Coverage:** 1990–2023
- **Completeness:** 80–97% depending on indicator

Data Cleaning Process

- Removed empty columns and aggregate regions (e.g., “World”, “OECD”).
- Standardized income group categories (High, Upper Middle, Lower Middle, Low).
- Converted dataset from wide format (years as columns) to long format (year as rows).
- Filtered data to 1990–2023 for consistency.
- Added calculated columns for Theme, Region, Income Group, and Decade.
- Excluded forecast years (2024–2025).

Analysis Methodology

- **Modeling:** Star Schema with fact table (measurements) and dimension tables (country, indicators, date).
- **Tools:** Power BI, Power Query, DAX measures.
- **Visualizations:** Line charts, bar charts, maps, scatter plots.
- **KPIs:** GDP Growth Rate %, GDP per Capita PPP, Internet Users %, Mobile Subs per 100, Broadband per 100, Countries in Recession.

ANALYTICAL USE CASES

A. GDP Growth & Economic Performance

Key Findings

- Global GDP growth averaged 2.5%, with 204 countries experiencing recession at some point.
- The United States (21.77K GDP Billion USD) and China (14.1K GDP Billion USD) dominate global output.

- Smaller economies often grow faster, but regional context (e.g., East Asia vs Sub-Saharan Africa) influences growth more than size.

Insights

- High-income countries show stable but slower growth.
- Lower-income countries demonstrate volatility but higher growth potential.
- GDP per capita PPP reveals inequality. Large economies are not always the richest per person.

Recommendations

- Policymakers should focus on inclusive growth strategies that balance GDP expansion with per capita improvements.
- Development agencies should monitor countries in recession as early warning signals for global instability.

B. Technology & Internet Adoption

Key Findings

- Internet users grew 3067.8% between 1990–2023.
- High-income countries lead with 4846.97% internet adoption, while low-income countries lag at 556.05%.
- Broadband penetration remains highly unequal (21.36 per 100 in high-income vs 0.31 in low-income).

Insights

- The digital divide persists across income groups, limiting opportunities for low-income countries.
- Mobile subscriptions are widespread, but broadband access is the true differentiator of digital quality.

- Regions with higher internet penetration tend to show stronger social and economic development indicators.

Recommendations

- Governments in low-income regions should prioritize broadband infrastructure.
- Mobile-first strategies can serve as transitional solutions where broadband rollout is slow.
- International organizations should support digital literacy programs alongside infrastructure investment.

CONCLUSION

The WDI Dashboard highlights two critical aspects of global development:

- a) **Economic Growth:** While large economies dominate global output, smaller and lower-income countries show higher growth potential but face volatility.
- b) **Technology Adoption:** Digital connectivity has expanded dramatically, but inequalities remain stark, especially in broadband access.

For sustainable development, bridging the digital divide and ensuring inclusive economic growth are essential priorities.

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